STUDENT ACTIVITY

Data Science Libraries — Activity

Activity: Data Pipeline Sequencing and Library Mapping

Q: Read the four unordered steps of a data analysis workflow below. Order them logically from raw numerical processing to advanced graphical representation. For each step, identify the primary Python library (NumPy, Pandas, Matplotlib, or Seaborn) that is specifically designed to handle that task.
Next, briefly explain the relationship and difference between Matplotlib and Seaborn in two sentences.

The Workflow Steps (Unordered):

- **Step A:** Generating a highly attractive, advanced statistical distribution plot of the final dataset.
- **Step B:** Executing fast, multi-dimensional array multiplication for initial numeric processing.
- **Step C:** Creating a basic line chart to quickly check the trend of the data.
- **Step D:** Creating a two-dimensional tabular data structure to clean the data and filter out missing values.

1. Arrange the Workflow Steps in the Correct Order and Identify the Python Library

Correct Order	Workflow Step	Python Library	Reason
1	**Step B:** Executing fast, multi-dimensional array multiplication for initial numeric processing.	**NumPy**	NumPy provides high-performance multi-dimensional arrays and mathematical operations, making it ideal for numerical computations.
2	**Step D:** Creating a two-dimensional tabular data structure to clean the data and filter out missing values.	**Pandas**	Pandas offers DataFrames for organizing, cleaning, filtering, and analyzing structured data efficiently.
3	**Step C:** Creating a basic line chart to quickly check the trend of the data.	**Matplotlib**	Matplotlib is the standard Python library used to create basic charts such as line, bar, scatter, and pie charts.
4	**Step A:** Generating a highly attractive, advanced statistical distribution plot of the final dataset.	**Seaborn**	Seaborn is built on Matplotlib and is specifically designed for creating attractive, advanced statistical visualizations with minimal code.

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Correct Workflow Sequence

Raw Numerical Processing
↓
NumPy
↓
Data Cleaning & Filtering
↓
Pandas
↓
Basic Data Visualization
↓
Matplotlib
↓
Advanced Statistical Visualization
↓
Seaborn

2. Explanation of Each Library

1. NumPy

- Used for numerical computing.
- Supports fast multi-dimensional arrays.
- Performs mathematical operations such as matrix multiplication, linear algebra, and statistical calculations.

```
import numpy as np
A = np.array([[2, 3], [4, 5]])
B = np.array([[1, 2], [3, 4]])
print(np.dot(A, B))
```

2. Pandas

- Used for data manipulation and analysis.
- Provides DataFrame and Series data structures.
- Helps clean data, remove duplicates, and handle missing values.

Example

```
import pandas as pd
df = pd.read_csv("students.csv")
```

df = df.dropna()

3. Matplotlib

- Used for creating basic graphs and charts.
- Suitable for line charts, bar charts, scatter plots, histograms, and pie charts.
- Provides complete control over chart customization.

Example:

```
import matplotlib.pyplot as plt
plt.plot([1,2,3],[5,7,9])
plt.show()Example
```

4. Seaborn

- Built on top of Matplotlib.
- Creates beautiful statistical graphics with less code.
- Used for heatmaps, box plots, violin plots, pair plots, and distribution plots.

Example

```
import seaborn as sns
sns.histplot(data=df, x="Marks")
```

3. Relationship and Difference Between Matplotlib and Seaborn (2 Sentences)

Matplotlib is the foundational Python library for creating a wide variety of basic and highly customizable plots, while Seaborn is built on top of Matplotlib and simplifies the creation of attractive, statistical visualizations. Seaborn provides better default themes and specialized statistical plots with less code, whereas Matplotlib offers greater flexibility and fine-grained control over plot customization.

Order	Step	Library
1	Initial numeric processing (multi-dimensional array multiplication)	**NumPy**
2	Data cleaning and filtering (tabular data)	**Pandas**
3	Basic line chart	**Matplotlib**
4	Advanced statistical distribution plot	**Seaborn**

Relationship & Difference

- **Relationship:** Seaborn is built on top of Matplotlib and uses it as the underlying plotting engine.

- **Difference:** Matplotlib is best for creating highly customizable general-purpose plots, while Seaborn focuses on creating visually appealing statistical graphics with simpler code.